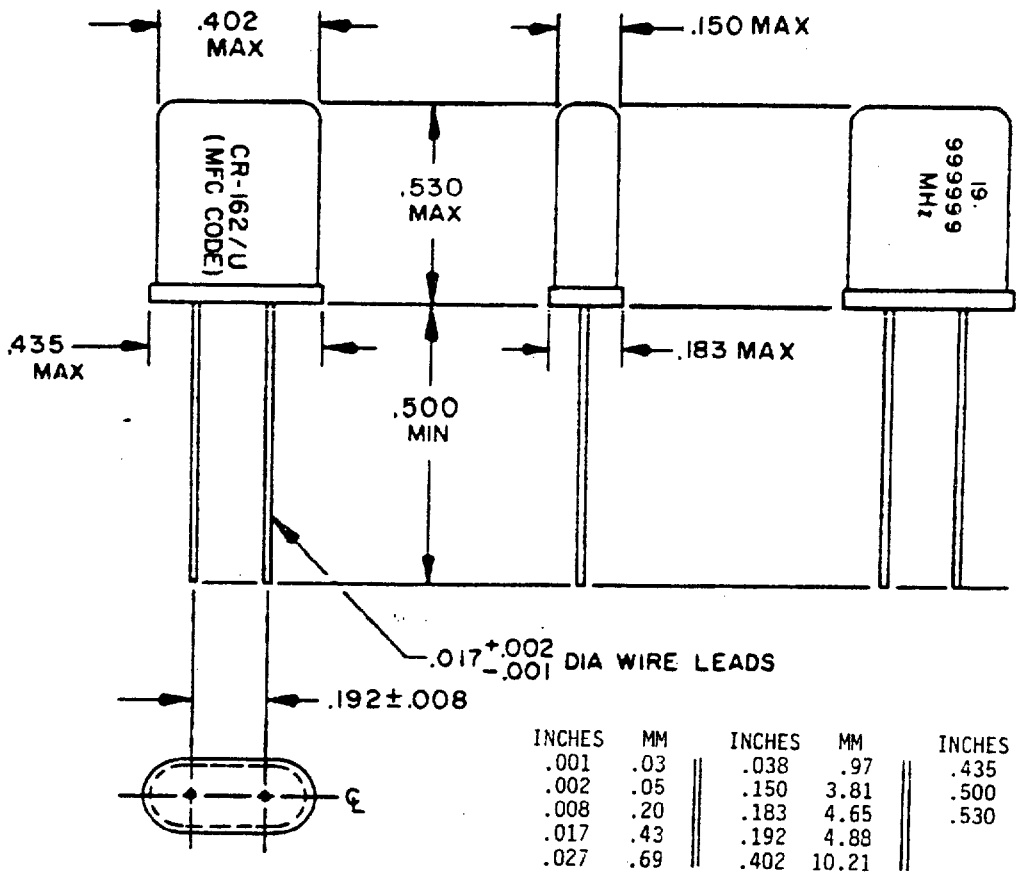
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-162/U

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 2.9 to 20 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.038	.97	.435	11.05
.002	.05	.150	3.81	.500	12.70
.008	.20	.183	4.65	.530	13.46
.017	.43	.192	4.88		
.027	.69	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 inch in height and not less than .027 inch in width.

FIGURE 1. Crystal unit, quartz, CR-162/u.

Ⓐ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-43/U (MIL-H-10056/30)
Dimensions, marking, and configuration	See figure 1
Frequency range	2.9 to 20 MHz, inclusive
Equivalent resistance	3.7 pF, minimum
Mode of oscillation	See table III
Test set, calibration values, and rated drive level	Fundamental
Frequency tolerance	See table II
A	±50 ppm
B	±100 ppm
Antiresonance, load capacitance	30.0 ±0.5 pF
Operating temperature range	
A	-55°C to +105°C, inclusive
B	-55°C to +135°C, inclusive
Capacitance, shunt	7.0 pF, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values		Resistor voltage drop	Rated drive level
		Resistance	Crystal current		
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	2.9 to 5.1	50	10	----	5.0 ±1.0
	5.1+ to 7.5	22	15	----	5.0 ±1.0
	7.5+ to 10.0	13	20	----	5.0 ±1.0
TS-683/TSM	10+ to 20	10	--	0.22	5.0 ±1.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.9 to 3.75	180
3.75+ to 4.75	120
4.75+ to 6.0	75
6+ to 7	50
7+ to 10	30
10+ to 20	25

Ⓐ QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE: This crystal unit is identical to the CR-64/U, except for the high temperature rating of 135°C and the holder. This crystal unit may be used to replace the CR-64/U.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17, 19

Preparing activity:
Air Force - 85

(Project 5955-F186-11)

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